INCH-POUND

MIL-M-14/4
2 April 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, POLYESTER, THERMOSETTING, CONTAINING MINERAL/GLASS FILLERS

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-14.

Polyester resin molding compounds, mineral/glass filled shall be of the following types, as specified:

Type MAG	–	Mineral filler, general purpose
Type MAI-30	–	Mineral filler, glass-fiber reinforced; nominal-impact strength, 3.0 foot-pounds per inch notch
Type MAI-60	–	Glass fiber filler, impact-resistant; nominal-impact strength, 6.0 foot-pounds per inch notch
Type MAT-30	–	Glass fiber reinforced, mineral filler, impact-resistant, track- and flame-resistant; nominal impact strength, 3.0 foot-pounds per inch notch.

REQUIREMENTS

Qualification test requirements are specified in table I. Batch acceptant test requirements are specified in table II.

NOTES

Type MAG. This type is a mineral-filled, polyester compound for use where good dielectric properties and arc resistance are required.

Type MAI-30. This type is a mineral-filled, glass-fiber-reinforced, polyester resin molding compound. It is an arc-resistant, flame-resistant, heat-resistant, high-impact compound having good mechanical and excellent electrical characteristics.

AMSC N/A FSC 9330

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Type MAI-60. This type is a glass-fiber-filled polyester compound for use where high-impact strength, good dielectric properties, and arc resistance are required.

Type MAT-30. This type is a heat-resistant, track-resistant, flame-resistant, high-impact, mineral-filled, glass-fiber-reinforced polyester compound.

TABLE I. *Qualification test requirements for polyester resin molding compounds, mineral/glass filled.*

Requirement	Type MAG	Type MAI-30	Type MAI-60	Type MAT-30
		Mechanical/physical		
Compressive strength, endwise	15,000	20,000	18,000	18,000
Dimensional stability	0.2	–	–	–
Flexural strength	7,500	14,000	12,000	15,000
Heat deflection temperature	175	200	200	225
Heat resistance	130	155	155	155
Impact strength, side[1]	–	3.0	6.0	3.0
Tensile strength	3,500	6,000	3,500	4,000
Water absorption	0.5	0.5	1.5	0.5
		Electrical		
Arc resistance	175	160	130	180
Comparative track index	–	–	–	180
Dielectric breakdown:				
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)				
Step-by-step-test	40	45	40	40
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)				
Step-by-step test	35	45	35	40
Dielectric constant:				
at 1 kilohertz	6.2	6.3	6.0	6.0
	6.5	6.4	7.0	6.0
at 1 megahertz	5.7	6.2	5.7	5.5
	6.0	6.4	6.0	5.5
Dielectric strength:				
Short-time test	375	300	150	325
Step-by-step test	325	250	125	275
Short-time test	300	275	100	275
Step-by-step test	250	250	50	225

See footnote at end of table.

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TABLE I. *Qualification test requirements for melamine resin molding compounds, cellulose, mineral/glass filled – Continued.*

Requirement	Type MAG	Type MAI-30	Type MAI-60	Type MAT-30
		Electrical *(Continued)*		
Dissipation factor:				
at 1 kilohertz	0.04	0.02	0.03	0.03
	0.06	0.02	0.08	0.05
at 1 megahertz	0.02	0.01	0.03	0.03
	0.03	0.02	0.05	0.05
		Combustion		
Flame resistance:				
Ignition time	60	100	90	100
Burning time	200	75	90	50
Flammability	V-1	V-1	V-1	V-1
Toxicity when heated:				
Carbon dioxide	15,000	15,000	15,000	15,000
Carbon monoxide	1,000	1,000	1,000	1,000
Ammonia	2,500	2,500	2,500	1,000
Aldehydes as HCHO	50	50	50	50
Cyanide as HCN	60	60	60	60
Oxide of nitrogen at NO_2	100	100	100	100
Hydrogen chloride	100	100	100	100

[1]The side of the test specimen is that area formed by the chase of the mold.

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TABLE II. *Batch acceptance test requirements for polyester molding compounds, mineral/glass filled.*

Property to be tested	Type MAG	Type MAI-30	Type MAI-60	Type MAT-30
Arc resistance	175	160	130	180
Comparative track index	–	–	–	180
Dielectric constant at 1 megahertz	6.0	–	–	–
Dielectric strength, step-by-step	250	250	50	225
Dissipation factor at 1 megahertz	0.03	–	–	–
Flexural strength	7,500	14,000	12,000	15,000
Impact strength, side[1]	–	3.0	6.0	3.0
Water absorption	0.5	0.5	1.5	0.5

[1]The side of the test specimen is that area formed by the chase of the mold.

Custodians:
Army – ME
Navy – SH
Air Force – 20

Review activities:
Army – ER, MI, MR, AR, EA
Navy – AS, OS
DLA – GS

User activities:
Navy – YD, EC

Preparing activity:
Navy - SH
(Project 9330-1097-05)

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